

PROJECT FALCON

Wednesday Presentation Guide

Clear slide-by-slide script · Target duration: 9–11 minutes

Main message: Project FALCON is an affordable, local-first coastal monitoring buoy concept with a working software foundation. The funding request is intended to build, calibrate, and field-test the physical system. Do not claim that its wave prediction or sensors are already field validated.

Before You Present

- Assign one main speaker and one member who will operate the slides and dashboard.
- Speak slowly. Pause after every major idea instead of reading the slide word for word.
- Use the slide as visual evidence; use this guide to explain what the audience should understand.
- If interrupted by a question, answer briefly and return using the transition line.

Slide-by-Slide Script

Slide 1 — Project FALCON (35–45 seconds)

Goal: Introduce the team, the project, and the purpose of the presentation.

“Good day, everyone. We are the Project FALCON Research Group from Fullbright College. FALCON means an AI-powered Live Coastal Observation Network. It is our proposed affordable and solar-powered smart buoy for collecting localized coastal information. Today, we will show the problem we want to address, how the system is designed, what we have already developed, and what support we need to build and validate the physical prototype.”

Transition: “To understand why we designed FALCON, let us first look at the monitoring gap.”

Slide 2 — Why This Matters (50–60 seconds)

Goal: Explain the problem without criticizing PAGASA or claiming that no coastal information exists.

“Official forecasts are important, and FALCON is not intended to replace them. The gap we are studying is affordable and site-specific observation. A wide-area forecast may not describe the exact condition at one near-shore research site, port, or small-boat launch area. Professional oceanographic equipment may also be difficult for a small school or local organization to purchase and maintain. Our research question is whether a modular and local-first buoy can provide useful observations at one defined Palawan pilot site at a manageable cost and complexity.”

Transition: “Our response to this gap is a small monitoring system that can measure, process, store, and explain data locally.”

Slide 3 — Proposed Solution (55–65 seconds)

Point to Measure, Process Locally, Display and Alert, and Research Prediction in that order.

“FALCON works in four simple stages. First, its sensors measure buoy motion, water pressure, wind, GPS position, temperature, battery, and solar condition. Second, the ESP32 repeatedly reads the sensors, while the Orange Pi stores and processes the information and hosts the local dashboard. Third, users can view current readings, history, equipment health, and abnormal-displacement alerts. Finally, after enough real field data are collected, we will train and evaluate a short-horizon wave-height model. The prediction shown today is a research feature and simulated demonstration, not yet a validated forecast.”

Transition: “The next image shows where these functions are located on the proposed buoy.”

Slide 4 — Upper Assembly and Sensor Levels (55–65 seconds)

Point from the top sensor deck downward. Do not discuss an underwater pointed section because it is not part of the approved visual.

“This image focuses on the above-water assembly. At the top, the wind sensor, GPS, and antennas need clear exposure. Below them, the solar panels are attached to a cross-braced tower, with cables routed through protected paths. The large upper electronics pod is the dry service area for the Orange Pi, ESP32, storage, interface boards, and power monitors. The BNO085 motion sensor should be fixed to the rigid central structure so its movement represents the buoy and not a loose panel. At the bottom is the cylindrical main float, whose waterline and freeboard still need to be verified through physical testing.”

Transition: “Inside that structure, the sensors and power system follow this proposed electronics plan.”

Slide 5 — Hardware Concept (60–70 seconds)

Explain function, not every wire. Point first to sensors, then ESP32, Orange Pi, and finally the protected power chain.

“Each component has a specific responsibility. The BNO085 measures orientation and motion, while the Bar02 measures water pressure that may assist wave estimation after calibration. GPS provides position, the weather sensor measures wind, two INA260 boards monitor battery and solar branches, MCP9808 measures enclosure temperature, and ADS1115 reads wind direction. The ESP32 gathers these sensor readings and sends telemetry to the Orange Pi, which handles storage, processing, the API, and dashboard. The lower part shows the proposed protected path from solar charging to the

battery and separate regulated supplies. This diagram is a design guide; final wiring will follow the purchased-board datasheets, measurements, fusing calculations, and approved pinout.”

Transition: “To make this system practical, the electronics must also be organized for safe maintenance.”

Slide 6 — Maintainable Enclosure (50–60 seconds)

Point to the edge-computing zone, sensor-control zone, terminal area, then cable glands.

“The upper box is designed as a serviceable electronics pod. A removable tray separates the Orange Pi computing area from the ESP32 and sensor-control area. Terminal blocks organize the regulated power, ground, and signal connections so technicians do not need to trace loose wires. Every external cable should pass through a labeled locking gland, include strain relief, and carry the same identifier at both ends. Power and signal wiring are separated to reduce interference. The final arrangement still requires heat, waterproofing, electromagnetic-compatibility, and maintenance-clearance tests.”

Transition: “With the hardware organized, this is how information travels through the system.”

Slide 7 — System Workflow (45–55 seconds)

Trace the arrows from sensors to ESP32, Orange Pi, database/API, then dashboard.

“The workflow is straightforward. Sensors take measurements, and the ESP32 checks and packages them into a consistent telemetry format. A USB or validated UART connection sends the data to the Orange Pi. The Orange Pi stores the records locally, performs signal processing, evaluates alerts, serves the API, and later runs the validated prediction model. The dashboard converts these records into understandable values, graphs, history, and system-health information. Because the main workflow is local, continuous cloud access is not required.”

Transition: “We have already implemented the software foundation for this workflow.”

Slide 8 — Current Progress (50–60 seconds)

Emphasize both the completed work and the honest limitation shown in the red/simulation notice.

“FALCON is more than an illustration. We already have a responsive dashboard, ESP32 firmware foundation, telemetry specification, sensor and fault simulator, Python edge service, local database, alert logic, wiring plan, bill of materials, and technical documentation. The firmware, backend tests, and dashboard build have been checked in software. However, the values on the dashboard are simulated and are not field measurements. Physical sensor integration, calibration, marine testing, and field-trained AI remain the next development stage.”

Transition: “This clear separation between completed work and future validation is also part of our innovation approach.”

Slide 9 — Innovation (45–55 seconds)

Do not say that the team invented the buoy, AI, or sensors.

“Our innovation claim is responsible system integration, not the invention of every component. FALCON combines affordable and replaceable hardware, local data collection, edge computing, solar-energy monitoring, a human-readable dashboard, and a prediction research workflow. It is local-first, so essential monitoring and storage are not dependent on a cloud connection. The interface also separates measured, simulated, estimated, and predicted information so users are not misled. The research contribution will be the documented and validated integration of these parts for one defined local coastal use case.”

Transition: “The value of that integration depends on who will use it and what specific need it will serve.”

Slide 10 — Beneficiaries (40–50 seconds)

Name possible beneficiaries, then narrow the study to one primary partner.

“Possible beneficiaries include a coastal LGU or DRRM office, fisherfolk and small-craft users, schools and researchers, and later ports, tourism groups, or environmental organizations. But we will not claim that one prototype immediately serves all of them. Our first step is to confirm one primary partner, one accessible Palawan pilot site, and one realistic information need. This keeps the research measurable and prevents the project from becoming too broad. If the first pilot demonstrates value, the system can later be adapted or expanded.”

Transition: “To reach that verified pilot, we propose a staged development plan.”

Slide 11 — Development Plan (45–55 seconds)

Move through the roadmap from design and procurement to controlled testing and reporting.

“We estimate eight to twelve months for development. We will first finalize component choices, mechanical details, safety requirements, and procurement. Next, we will integrate the electronics, fabricate and seal the buoy, and calibrate every sensor under controlled conditions. Only after passing those checks will we conduct a supervised near-shore pilot. The collected data will then be used to compare simple prediction baselines with a lightweight model. The final stage is evaluation and documentation, including limitations and recommendations.”

Transition: “Completing these stages requires support for more than electronics alone.”

Slide 12 — Preliminary Budget (45–55 seconds)

Point to the categories instead of reading every amount. Call the amount a planning range.

“Our preliminary development estimate ranges from seventy-two thousand to one hundred twenty-seven thousand pesos. This covers sensors and controllers, the Orange Pi, solar and battery power, the marine structure and enclosure, mooring and safety materials, calibration, supervised field work, spare parts, transport, and contingency. The range represents the complete prototype-development process, not only the visible electronics. It is still a planning estimate and will be replaced by current supplier quotations and approved specifications before procurement. Any final spending will be documented through a bill of materials and test records.”

Transition: “With this support, these are the concrete results we intend to deliver.”

Slide 13 — Expected Result and Request (50–60 seconds)

End with the deliverables, specific request, and evidence-focused closing statement.

“At the end of funded development, we aim to deliver one integrated physical prototype, calibrated sensors and power system, a working local dashboard and data archive, supervised coastal-test results, and a comparison between the prediction model and a simple baseline. We will also provide reproducible technical documentation so the design, results, and limitations can be reviewed. We are requesting support for component procurement, fabrication, calibration access, supervised field testing, and technical mentorship. We are not asking you to accept an untested forecast. We are asking for the opportunity to build the prototype and generate credible local evidence. Thank you, and we are ready for your questions.”

Likely Questions and Simple Answers

Question	Suggested answer
Is the AI prediction already working?	“The interface and forecast pipeline work with simulated data, but the model is not yet field validated. We need calibrated local measurements before training and comparing it with a persistence baseline.”
What makes FALCON different?	“Its contribution is the affordable, local-first, and transparently validated integration of sensing, edge computing, power monitoring, prediction research, and displacement alerts for one defined Palawan use case.”
Will it replace PAGASA?	“No. FALCON is a research and localized observation tool. Official advisories and warnings remain the authority.”
Why use both ESP32 and Orange Pi?	“The ESP32 handles reliable sensor reading. The Orange Pi handles storage, processing, the dashboard, and future model inference. Separating these roles

improves reliability and maintainability.”

How will you prove the wave measurements are accurate? “We will calibrate the sensors, document the buoy and mooring response, and compare synchronized results with a suitable reference such as a pressure logger, staff gauge, video method, or validated wave instrument.”

How will it operate without internet? “Measurements are processed and stored locally. Internet or long-range communication can be used for remote access, but the core acquisition and archive do not depend on continuous cloud service.”

Why is the budget a range? “The exact marine-rated components, fabrication method, and suppliers still require final quotation. The range prevents us from presenting an invented exact amount before procurement.”

What happens after funding? “We finalize the design, purchase verified components, assemble and calibrate the system, perform controlled tests, conduct a supervised pilot, evaluate the model, and submit documented results.”

Final Reminder

If you forget a line, return to this sentence: **“FALCON is a working software concept that needs funded physical development, calibration, and supervised field validation.”**